to overspending on project, project overrun, loss of reputation or failure to deliver product on time. The most inevitable part of software development is deployment. Automation can be used or implemented throughout the entire lifecycle. A process which needs no human intervention can be automated to avoid expensive resources doing repetitive and mechanical tasks so that end goal can be achieved quickly, accurately and consistently. Development workflow should include deployments as critical part. Workflow usually includes at least 3 environments:

1. Development
2. Staging
3. Production

Developers works on their own separate branch for developing any feature or to fix any bug. Minor fixes can be directly fixed on stable development branch (trunk). Once the feature is implemented, the code is merged into the staging branch so that it can be deployed on staging server for testing and quality assurance. After testing phase, feature code is merged into development branch. Then on releasing new feature the feature branch is then merged to the production so that new feature can be deployed on production servers.

Testing is also a crucial part of deployment lifecycle

Development Environment:

Every developer has their own local setup for working on fixes or features. Some companies have development environments configured for automatic build deployment for every commit or push or whenever required. This